interface (8251). 3½+3½

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5. (a) Write any two assembler directives for 8085 microprocessor. 2

(b) Describe about any two subsystem available that aid a hardware designers in building his own system arround the subsystem. 3½+ 3½

(c) Write short notes on operating system. 7

(d) Write the functional description of system monitor & text editor. 7

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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(Electrical Engg. Branch)

MICROPROCESSORS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each questions is compulsory. Attempt any two parts from (b), (c) & (d) of each question.*

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1. (a) Why 8085 microprocessors can access only 64 k bytes of memory. 2

(b) Write the function of following pins of 8085 microprocessor.

(i) ALE 1

(ii) Ready 1

(iii) Trap 1

(iv) S_1 and S_0 1

(v) Reset out 1

(vi) SID 1

(vii) $IO/\overline{m}$ 1

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(c) What do you mean by addressing mode? Explain all addressing mode provided by 8085 microprocessor. 2+5

(d) Write a program to arrange 12 bytes of data in ascending order. Assume the data is stored in memory as an array starting from F500H onwards. 7

2. (a) Give the classifications of semiconductor memory. 2

(b) Draw the timing diagram for write cycle of static RAM HM 6264. 7

(c) Describe all input and output electrical signals common to most memory chips. 7

(d) Write short notes on shadow RAM. 7

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3. (a) What do you mean by absolute decode technique & partial decoding technique. 2

(b) Explain various data transfer schemes available for microprocessor based system. 7

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(c) Draw the neat schematic showing the interfacing between 8085 microprocessor and : 2+5

(i) L6 kB RAM, starting address 4000 H

(ii) 8 kB ROM, Assume suitable address

Use partial decode technique for decoder logic.

(d) Explain masking & reading of interrupts in 8085 microprocessor with example programs. 3½+3½

4. (a) Differentiate between I/O mapped I/O & memory mapped I/O technique to interface I/O devices to microprocessor. 2

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(b) Draw the neat schematic showing the interfacing between 8085 microprocessors and 3 slave programmable interrupt controller (PIC) connected to one master programmable interrupt controller (PIC). Use I/O mapped I/O technique to connect peripherals. 7

(c) Draw & explain the functional block diagram of direct memory access controller (8257). 7

(d) Explain mode word format for synchronous & asynchronous mode in programmable communication